Explainable AI with SHAP Values

Opening the Black Box of Machine Learning Models

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Abstract.
Although machine learning algorithms like Random Forest and Gradient Boosted Trees have high prediction accuracy, their reasoning capability is unclear. This is a major limitation in high-stakes industries like finance and medicine. In this lesson, you will learn SHAP (SHapley Additive exPlanations), a theoretically sound technique based on cooperative game theory that provides a precise and fair contribution of each feature to any prediction. SHAP will be used to produce feature summaries, waterfall charts for individual predictions, and dependence graphs that highlight feature interactions after training a Random Forest classifier on the Wisconsin Breast Cancer dataset. At the end of this lesson, you will be able to apply SHAP to your own tree-based models and critically understand the results. You will also be able to make informed decisions about using SHAP.

1.Introduction

While the advances made in the predictive capabilities of machine learning systems in the present day have been considerable, the price of this increased power is, in many instances, the concomitant lack of transparency in the decision-making process of the most accurate models available. The question of a clinician, for example, “Why does the machine learning model classify this tumor as malignant?” or of a regulator, “Why was the loan request denied?” can be answered by a random forest or a deep neural network with a high degree of accuracy, but the machine learning model will be unable to provide a clear rationale for its actions. The challenge of machine learning in the present day is the lack of transparency in the decision-making process of the most accurate models available.

In order to tackle this problem, the current lesson will cover SHAP, or SHapley Additive exPlanations, a method based on the theory of cooperative games, which ascribes to each feature a mathematically exact portion of the credit or blame for each individual prediction made by the model, without simplifying or replacing the model itself.

The Wisconsin Breast Cancer dataset, which is a well-known standard dataset with 30 numerical features based on the cell image data and the binary target variable for malignant or benign cancer, is used throughout the course. The dataset is used for training the Random Forest classifier, and the classifier is then further examined using the SHAP method.

2. Background: The Black-Box Problem

High-stakes decision-making in domains such as medicine and finance requires auditability. The European Union’s AI Act and Regulation (EU) 2016/679 (GDPR) Article 22 provides for a right to explanation for automated decision-making. SHAP differs from other heuristic methods in that it provides a principled game-theoretic framework for explanation that is both consistent and accurate.

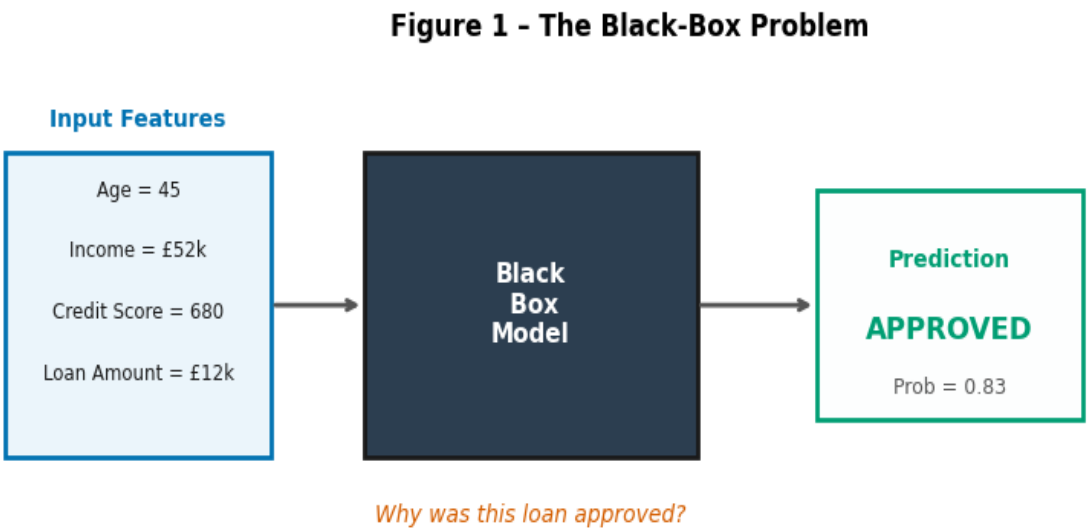

Figure 1 The conceptual diagram showing inputs going into a black box and an "Approved/Probability" output coming out

3. The Mathematics of SHAP Values

3.1 Shapley Values

The Shapley Values were developed by Lloyd Shapley in 1953 to address the problem of the "coalition game" or how to divide a payoff among the players of a game. In the context of Machine Learning, the payoff is the prediction, and the players are the features. The SHAP value of a feature, (ϕi), is the average marginal contribution of the feature over all possible combinations:

$$\varphi_i = \Sigma\ [|S|!(|F|-|S|-1)!\ /\ |F|!] \times [f(S \cup \{i\}) - f(S)]$$

3.2 Theoretical Guarantees

The SHAP approach is the only one that meets the four basic axioms:

- Efficiency: SHAP values add up to the exact difference between the prediction and the "average" base value.
- Symmetry: Contributors to the prediction that are the same will have the same SHAP value.
- Dummy: Features that do not contribute to the prediction have a SHAP value of zero.
- Consistency: If a feature's contribution to the prediction increases in another model, its SHAP value will not decrease.

3.3 TreeSHAP

The exact calculation of the Shapley value is computationally complex, with exponential time complexity of (2n). However, the TreeSHAP algorithm has improved the calculation of the SHAP values, especially for tree-based models like Random Forest and XGBoost, with a polynomial time complexity $O(TLD^2)$

Property / Axiom	Meaning
Efficiency (Local Accuracy)	SHAP values sum exactly to the difference between the model's prediction and the base value (average training prediction). Explanations are exact, not approximate.
Symmetry	Features with identical marginal contributions receive identical SHAP values, regardless of their position in the feature vector.
Dummy (Missingness)	A feature that has zero effect on any prediction receives a SHAP value of zero.
Consistency	If a model changes so that feature i contributes more, its SHAP value never decreases — unlike MDI and permutation importance.

4. Dataset and Model Training

The dataset used for the experiment is the Wisconsin Breast Cancer dataset, which has 569 instances and 30 numeric attributes (e.g., radius, texture, concavity). We use a Random Forest classifier to obtain a near-perfect level of discrimination (ROC AUC ≈ 0.997).

Code Block: Model Initialization

```
import shap
from sklearn.ensemble import RandomForestClassifier
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split

# Load data and train
data = load_breast_cancer()
X_train, X_test, y_train, y_test = train_test_split(
    data.data, data.target, test_size=0.2, random_state=42,
stratify=data.target)

rf = RandomForestClassifier(n_estimators=100, random_state=42)
rf.fit(X_train, y_train)

# Initialize TreeSHAP
explainer = shap.TreeExplainer(rf)
shap_values = explainer(X_test)
```

5. Global Interpretability

Global Interpretability helps us understand the overall logic of the model for the entire population.

5.1 Feature Importance

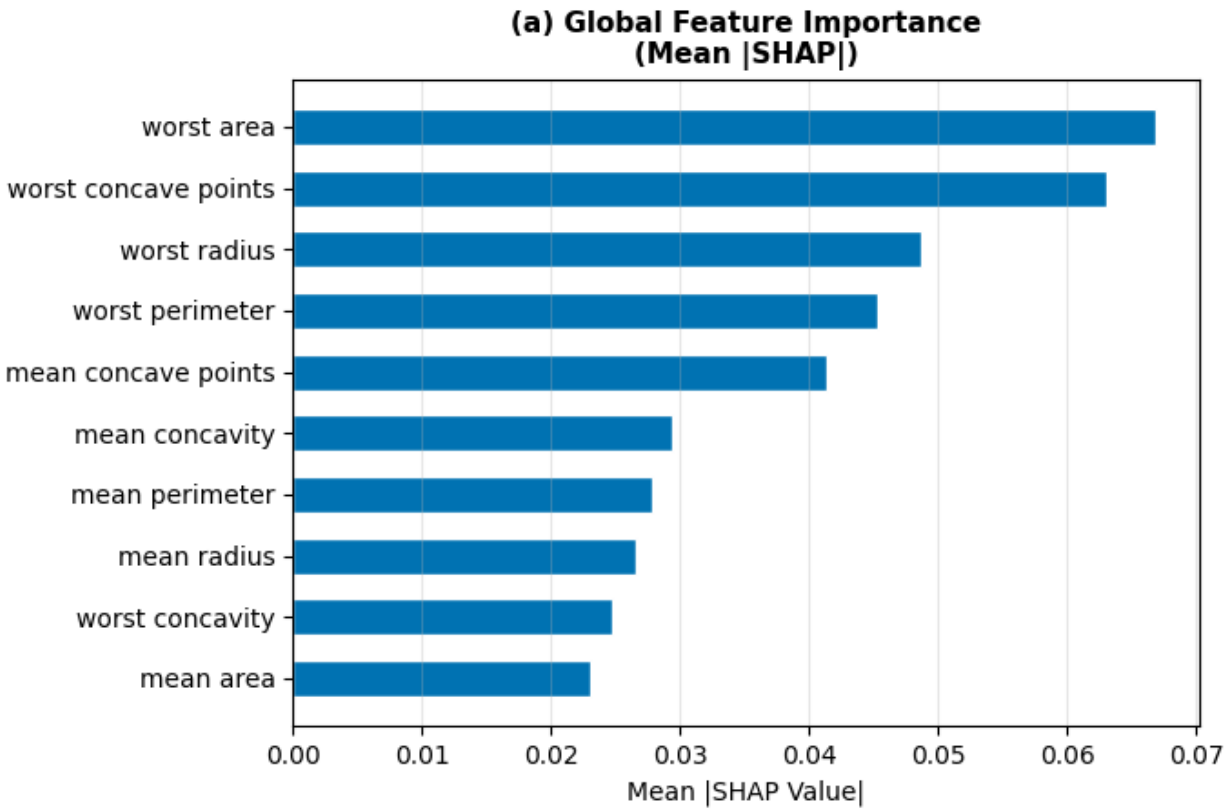

Figure 3a The horizontal bar chart showing "mean |SHAP value|"

A SHAP bar chart shows the feature importance in descending order of the mean absolute SHAP value for each feature. The SHAP values are in the same scale as the model output, which is probability in this case. If the worst radius feature has a SHAP value of 0.32, it changes the probability of malignancy by 32%.

5.2 Beeswarm Plots

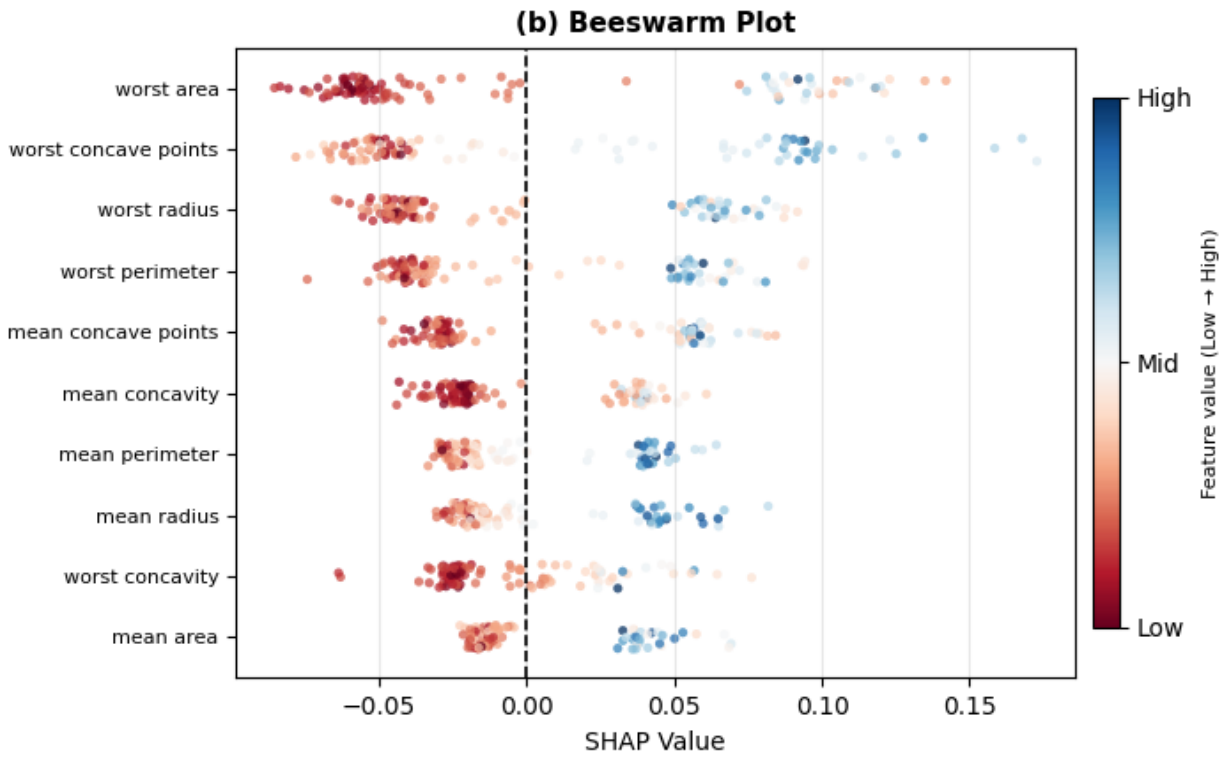

Figure 3b The scatter/beeswarm plot with red and blue dots

Beeswarm plots provide the overall distribution of the SHAP values, including the direction of the effects. The points on the graph represent the samples, and the color of the points indicates the feature value (Red = High, Blue = Low). The worst radius feature, which has high values (Red), generally has positive SHAP values, indicating a strong push for "Malignant" classification.

6. Local Interpretability: Per Prediction Explanations

6.1 Waterfall Plots

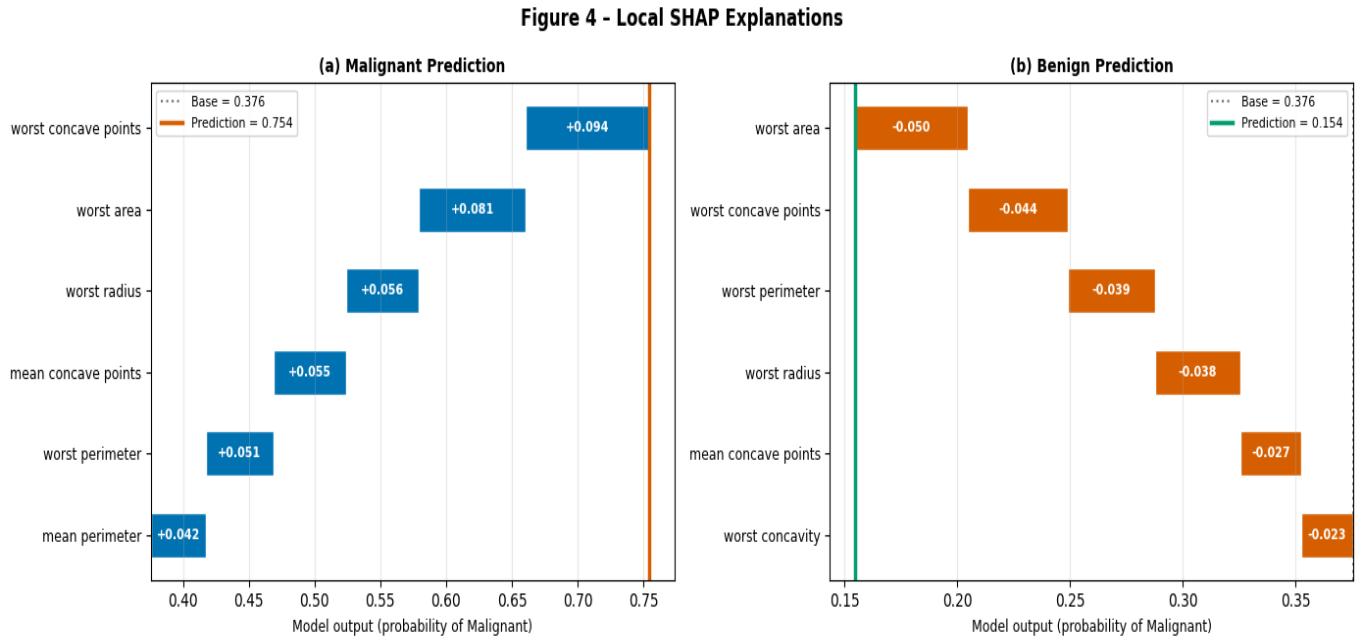

Figure 4 (The side-by-side waterfall charts for a "Malignant" and "Benign" case)

Waterfall plots provide explanations for individual patient results. The plots start at the "base value" (the average prediction of all training data) and show how each feature's SHAP value changes the prediction to arrive at the final result.

6.2 Numerical Verification

SHAP's Local Accuracy guarantee ensures that for any patient:

$$(\text{Base Value} + \sum \text{SHAP Values} = \text{Model Prediction})$$

7. Interactions and Comparisons

7.1 Feature Interactions

The dependence plots provided by SHAP account for non-linear relationships. For example, the worst radius may not on its own be very predictive of malignancy, but its high value only becomes significant when combined with a high value for the worst concave points.

7.2 SHAP vs. Alternatives

Method	Accuracy	Consistency	Local Explanations
MDI (Gini)	High	Low (Biased)	No
Permutation	Medium	Medium	No
SHAP	**High**	**High**	**Yes**

8. When to Use (and Avoid) SHAP

Use SHAP when	Be cautious when
Providing a regulatory audit trail.	Features are highly correlated (multicollinearity).
Debugging data leakage.	You need causal (not just associative) logic.

Use SHAP when	Be cautious when
Auditing for fairness/bias.	Using deep learning (requires approximations).

9. Debate and Mor al Implications

9.1 Advantages and Disadvantages

The major advantage of SHAP is its theoretical basis. SHAP can explain the model but not necessarily the reality. "If SHAP identifies that there is a 'spurious correlation' (e.g., that a hospital ID number is predictive of a disease), SHAP will correctly identify that the ID number is 'significant.' Human domain knowledge is always the final filter."

9.2 Ethical Artificial Intelligence

There is a danger of "explanation washing," where SHAP is used to "make an unfair model appear fair." Transparency is different from fairness. "While SHAP can be used to identify that there is high SHAP value for race or gender in a model where there shouldn't be, SHAP is a whistleblower. SHAP does nothing to eliminate bias."

Explanations must be "action-oriented" in a health environment. "If SHAP value is not useful to a doctor to assist a patient, then what is the point?"

10. Conclusion

SHAP helps to create a transparent and auditable machine learning system. SHAP's ability to create unique and fair attributions makes it an ideal choice for deploying AI in high-stakes fields. As AI becomes further integrated into society, its ability to explain its decision is just as important as its decision.

11. References

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